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A Hidden Threat?:
Investigating the Effects of Radiation on Life Expectancy in Japan

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Abstract

The effects of radiation on the health of a population have been studied meticulously since the emergence of nuclear technology in the past century and caused public concern. This study aims to analyze whether nuclear radiation has a strong effect on life expectancy in order to understand the long-lasting implications of radiation exposure. To do so, radiation levels and life expectancy statistics from 2010-2020 from varying prefectures in Japan will be dissected. I use data from national records and environmental databases to run a regression analysis on correlation of the variables and uncover their relationship. My findings showed that radiation did not have a significant effect on life expectancy, so I turn to exploring why that might have been the case, specifically in Fukushima prefecture, as it experienced the nuclear meltdown at Fukushima Dai-ichi nuclear power plant in 2011. Certain factors, such as evacuation, medical care, and community ties have been found to be important in the case of keeping residents in Fukushima safe. Overall, the study helps to gain perspective on the lasting effects of radiation and what actions may be able to curb its harmfulness.

Introduction

Radiation caused by mistakes and disasters with nuclear material, as brought about by advances in nuclear technology, is ever present in society today. Due to the use of nuclear materials and technology in war and country conflicts, as well as for energy production, radiation has an increased role in communities. The process of obtaining energy in this way risks the release of harmful nucleotides into the environment, which may include Cesium-137 and Cesium-134 which are the most common nuclear materials released in nuclear power plant accidents (*Ministry of the Environment*, n.d.). These radioactive materials affect the whole body

and can last in the environment for up to 30 years. Because of the known consequences of nuclear technology, there has been much debate between community members, governments, doctors, and the general public on whether or not it is safe to reside in areas where radiation exposure is high. Many are concerned whether exposure to radiation will negatively affect their lives in the future.

Japan has faced three large nuclear disasters: Hiroshima, Nagasaki, and Fukushima. Hiroshima and Nagasaki experienced the first and last atomic bombs ever used, and Fukushima prefecture was victim of the nuclear meltdown of Fukushima Dai-ichi nuclear power plant as a result of the Great East Japan Earthquake in March, 2011. Examining the patterns of radiation and life expectancy in prefectures of Japan will provide some insight on whether radiation exposure is causing serious medical problems and harm that result in a shorter lifespan for individuals. Analyzing life expectancy will provide a more holistic overview of the effect of nuclear radiation on people.

I hypothesize that exposure to radiation caused by the atomic bombings and nuclear meltdown in Japan will cause the life expectancy in associated prefectures to be lower because of the possibility of various health issues due to radiation. The effects of radiation can have a lot of impact on the human body and even though decontamination processes are ongoing, they might not be enough to offset the original exposure to radiation. Some may believe that radiation and life expectancy are positively correlated because of other factors such as evacuation-related injuries or death, or mental health issues and suicide. While these things do happen, they would not be explaining a long-lasting effect of radiation. When attempting to measure this relationship, it would be best to understand if illnesses that come from exposure to radiation and are more long term would affect life expectancy.

As I will discuss, there are many health concerns regarding radiation exposure, including cancer. Additionally, there is an ongoing discourse regarding whether community members in nuclear radiation disaster affected areas should be more or less concerned with the issue. Measuring life expectancy is a good way to see the overall effect, essentially providing a more simple take on whether radiation is causing significant and long-lasting harm in Japan. After analyzing the data, we will explore why radiation has the impact on life expectancy that is found.

Nuclear Meltdown and Radiation

The meltdown of a nuclear reactor occurs when the reaction inside the reactor is not controlled, leading to an increase of temperature of the reactor fuel and eventually melting of the fuel (Suppes & Storvick, 2007). If the temperature cannot be controlled it often leads to other issues inside the reactor which then, coupled with the melted fuel, creates a larger disaster. What was exhibited in both the Chernobyl meltdown was that the heat invaded the areas of the reactor that had water. Water was used as a cooling mechanism, but the heat from melting fuel ended up heating the water and increasing its pressure. This eventually led to hydrogen explosions in the plant because the fuel reacted with the water once it came into contact. The contact of fuel and water creates hydrogen, which builds up until it explodes the reactor. Once the explosion occurs, radiation is exposed to the environment outside of the reactor. In the case of Fukushima Dai-ichi, water from the tsunami caused the emergency energy equipment to fail, causing the water in the reactors to evaporate and dropping water levels. The fuel rods were then exposed to the water and vapor which caused the hydrogen explosions (*TEPCO*, n.d.).

Previous studies have worked towards uncovering the health issues related to nuclear radiation exposure. According to the Ministry of the Environment of Japan, radioactive materials

that are derived from nuclear disasters include H-3, Sr-90, I-131, Cs-134, Cs-137, and Pu-239 (*Ministry of the Environment*, n.d.). Out of these six radioactive materials, three of them (H-3, Cs-134, and Cs-137) are known to affect the entire body, while two affect bones (Sr-90 and Pu-239), and one affects the thyroid (I-313). Radiation from nuclear materials is known as ionizing radiation, and it can have very harmful effects because it has the ability to chemically alter the composition of a cell (ARPANSA, 2022). It can do this because of its ability to remove electrons from atoms, which is why it is used to produce nuclear energy in the first place. The United States National Center for Biotechnology Information states "a persistent increase in the rate of cell death has been shown to occur in cell populations many generations after radiation exposure" (Little, 2003). This would cause substantial concern to the communities affected by nuclear radiation because it suggests that radiation is very harmful as it is killing cells, and could even be interpreted as life threatening because it was seen in many generations.

The radioactive materials released with the highest frequency during the Fukushima disaster were I-131, Cs-134, Cs-137, and Sr-90. These materials have long half lives, meaning they stay in the environment for a considerable amount of time (from days to decades), and also in the body once they enter it, providing access to harm cells and DNA for anywhere from 10 days to 20 years. Interestingly, when comparing the Fukushima Dai-ichi meltdown to the nuclear meltdown that occurred in Chernobyl in 1986, Chernobyl experienced over twice the amount of Iodine-131, Cesium-134, and Cesium-137 into the atmosphere. Considering that nuclear meltdowns release materials that are damaging to the body and can cause harm for an extended period of time, it is natural for one to think that they would have a detrimental effect on human life, thus affecting those who experienced radiation from the Fukushima meltdown.

Health Concerns

Much of what is known about cancer related to nuclear disaster radiation exposure has been found from studies on the Chernobyl disaster in Ukraine. Children who were exposed to high radiation were found to be at higher risk for thyroid cancer than those who were not (National Cancer Institute, 2022). The radiation was found to also be able to cause cancer in the next generation, as people who experienced the radiation could have experienced genetic changes in sex cells, causing potential harm for their offspring. While this study focuses specifically on children, it still brings significant information that allows for concern about cancer amongst the general population. It supports the narrative that radiation does negatively affect the health of individuals, and therefore supports that it would be interesting to see if life expectancy was actually influenced by this.

Many people in Fukushima prefecture have expressed their fears and concerns about the ongoing higher levels of radiation in the area due to the nuclear meltdown. They understand the potential health risks of it and want to react accordingly to keep themselves and their families as safe as possible. One study conducted found that residents of Fukushima have long lasting anxiety regarding the radiation exposure (Nakayama et al., 2019). This was linked to the frequency and type of media they consumed, but nonetheless proved that anxiety is high amongst residents even years after the disaster. Additionally, the Fukushima Mother's Group has expressed health concerns with TEPCO's plan to release treated water from the power plant into the ocean (Takenaka et al., 2023). They worry that these efforts will remind the community of the initial disaster, creating physical and psychological issues for the community again, both being susceptible to water contamination and to fear, PTSD, and other mental health issues stemming from their experience over a decade ago. Both these examples serve to highlight the

ongoing worry in the community about whether or not they are really safe from the Fukushima nuclear meltdown, even many years later. This fear and anxiety amongst the public may drive how they reacted to the disaster. Someone who felt very scared of the risk of radiation may have decided to evacuate much earlier, while another who was not fearful decided to stay longer or not move at all. I believe that those who held larger amounts of fear and anxiety would have left the affected areas sooner, thus lowering their exposure to radiation and not impacting their life expectancy.

Additionally, fear and anxiety caused by this disaster has also led to more serious physical problems too. A study on the psychological effects of the disaster found that depression rates among adult evacuees were much higher than the rest of Japan, at 14.6% in 2012 when the country average was just 3% (Maeda et al., 2018). The prevalence of depression can lead to other issues such as alcholism and suicide. The study also references research done on suicide rates following disasters in Japan, which found that the rates only increase after a major disaster (as opposed to a smaller one). There were 83 cases of disaster-related suicide in Fukushima prefecture in the 5 years following the disaster, which was more than Iwate and Miaygi prefectures. Therefore proving that the mental health issues people face because of the disaster can also lead to detrimental outcomes in physical health and wellbeing.

Government Narrative

Government officials say there is little reason that residents of Fukushima prefecture and other affected areas should still fear health effects as a result of the disaster. The Prime Minister's Office of Japan released a report in 2013 written by a scientist that specializes in the field of radiobiology. It states, "acute health effects which can be caused by exposures to ionizing

radiation have not been observed during the first years after the accident," confirming that no significant health issues due to radiation were seen following the disaster (*Prime Minister of Japan and His Cabinet*, 2013). The report did mention the potential damage of in-utero cells, but dismissed the generalized claim that fetuses were affected as few abnormalities were observed. Finally, the report makes the claim that the spike in cancer cases is due to the fact that so many people were screened, so it essentially was just an effect of more people being tested. Additionally, a doctor who has been studying the radiation in children from Fukushima in comparison to Chernobyl, found similar results– that the radiation exposure in Fukushima was much less than that in Chernobyl and the scale of screening in Fukushima led to higher rates of cancer. (Wingfield-Hayes, 2014). It is clear that government officials believe the work being done to decontaminate the affected areas has been successful. However, if this is the case, why are so many residents fearful of radiation and scared to return to their homes? The competing narrative on the safety in Fukushima prefecture is difficult to unwrap.

It's well known that nuclear technology has harmful potential with its ability to release dangerous nuclear materials into the environment. They have known health consequences, as examined after the Chernobyl disaster, and create fear in many people who have been affected by a nuclear disaster. However, the Japanese government claims that the situation is under control and has opened up towns for residency again with plans to continue decontamination and get more residents to return. Comparing the government's take on the safeness of affected areas, what is known about nuclear radiation, and the health concerns posed by the general public presents a question: does radiation actually present long-standing effects on life expectancy?

Methods

In order to measure the effect of radiation from nuclear fallout on life expectancy I gathered data for both average life expectancy and average radiation levels per year over a period of time. The life expectancy data was gathered from the Japan Ministry of Health, Labour, and Welfare (*Ministry of Health, Labour, and Welfare,* n.d.) for the years 2005, 2010, 2015, and 2020. This data was used to examine how life expectancy was growing before the 2011 Triple Disasters and afterwards. Comparing these trends to other prefectures can give us an insight onto the impact of 3/11. Data on radiation levels in prefectures of Japan was gathered from the Environmental Radiation Database (*Environmental Radiation Database – Environmental Radioactivity and Radiation in Japan*, n.d.). Radiation levels were collected from 2010-2024 to see radiation levels before the Fukushima disaster and the period after the disaster. Levels were collected in nanograys per hour (nGy/h) and converted to microsieverts (μSv). Additionally, this data was cross referenced with data from Safecast's radiation map to ensure that the information was accurate.

To analyze the data, the information was inputted into an excel spreadsheet that included prefecture, year, male life expectancy, female life expectancy, average life expectancy, radiation levels in nGy/h and radiation levels in μSv. Radiation data for the year 2005 was not collected, so that had to be left out of the analysis. The life expectancy data still starts in the year 2005, but the radiation data begins with the year 2010. From there, I created graphs comparing average life expectancy and radiation levels in μSv across prefectures for 2010, 2015, and 2020.

The visualizations were made using Python, where I created scatter plots to show the trends of life expectancy and radiation. A regression line was then implemented for each graph in order to better understand the relationship between the variables. Correlation and statistical

significance were also evaluated as a final measure of understanding the relationship. Additionally, the use of a bar chart aided in visualizing other information within the data, varying amongst prefectures.

Data Analysis

This section provides the visual representation of life expectancy and radiation levels from 2010-2020. Starting with the year 2010, I analyze the effect of radiation and explain the data. Each graph includes a regression line to demonstrate if there is a trend between the variables as well as arrows signifying where Fukushima prefecture and Hiroshima prefecture fall on the graph. Those two prefectures were selected to highlight due to their experience with nuclear radiation and the time difference between when they occurred.

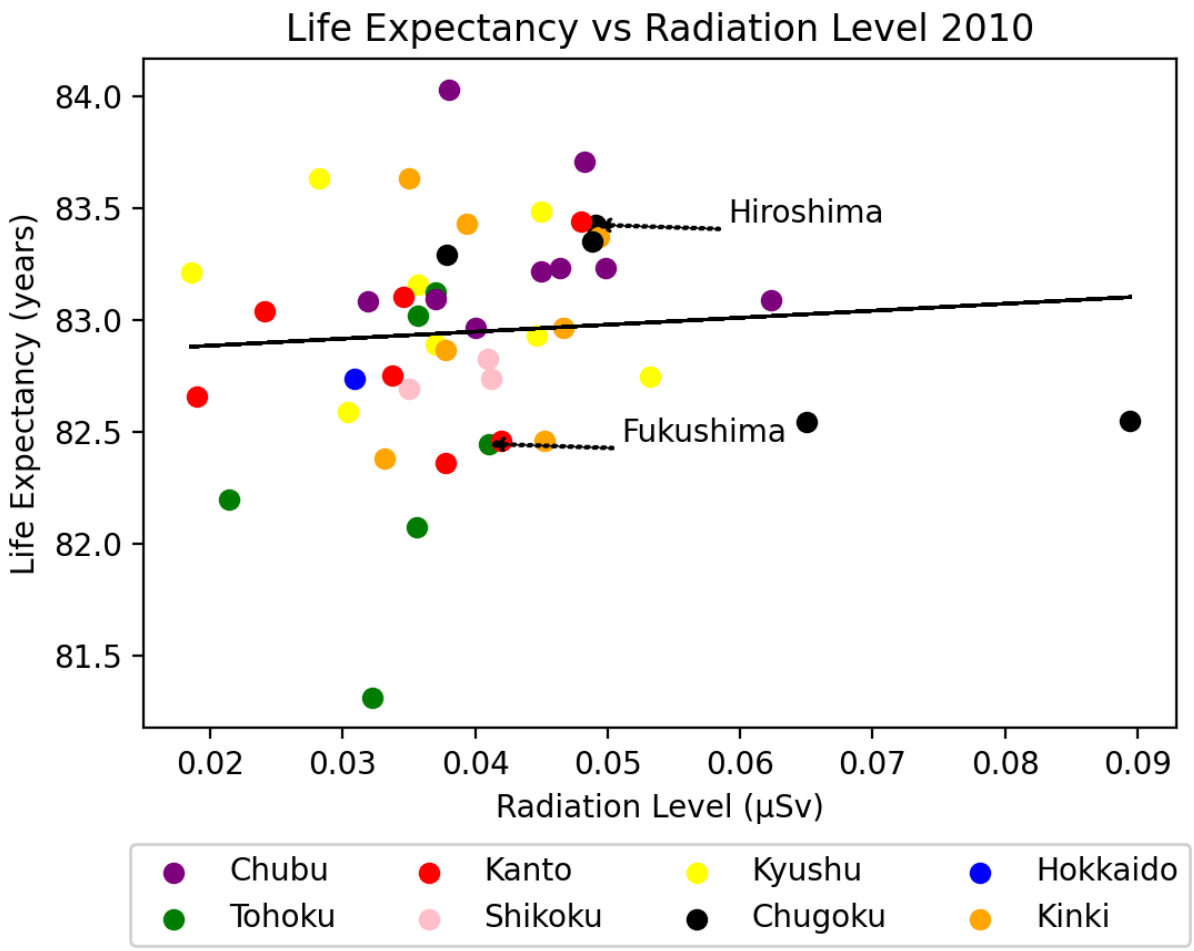


Figure 1: Demonstrates the relationship between life expectancy and radiation levels across prefectures in the year 2010. Graph is color coded by regions of Japan, while each individual dot is a prefecture. Includes a regression line and labels for specified prefectures.

The first figure focuses on radiation and life expectancy for 2010. This data is before the Fukushima nuclear meltdown, as that occurred the following year. It is important to analyze data prior to the disaster to understand how it changed afterwards. The graph shows there is no strong correlation between the two variables. However, the regression line is showing a slight positive trend. When coding for correlation and statistical significance, the results are as follows:

Correlation coefficient: 0.0773878330466356

T-statistic: 0.5206948962017873

P-value: 0.6051321133421577

The correlation is not significant at the 0.05 level.

The results show the correlation was 0.08, however the analysis was not statistically significant as the p-value was 0.60. There is much variation in life expectancy and radiation levels across prefectures, even across regions of Japan, which the graph is color coded by. In 2010, Fukushima had a life expectancy of 82.45 years and radiation level of 0.04 μSv. At the same time, Hiroshima had a life expectancy of 83.43 years and radiation levels of 0.05 μSv. Interestingly, Hiroshima had higher radiation levels for the year but also higher life expectancy. However, because the difference in radiation is so miniscule, I argue that it is insignificant. It does however point out how Hiroshima prefecture had returned to relatively normal radiation levels by 2010, after the disaster in 1945.

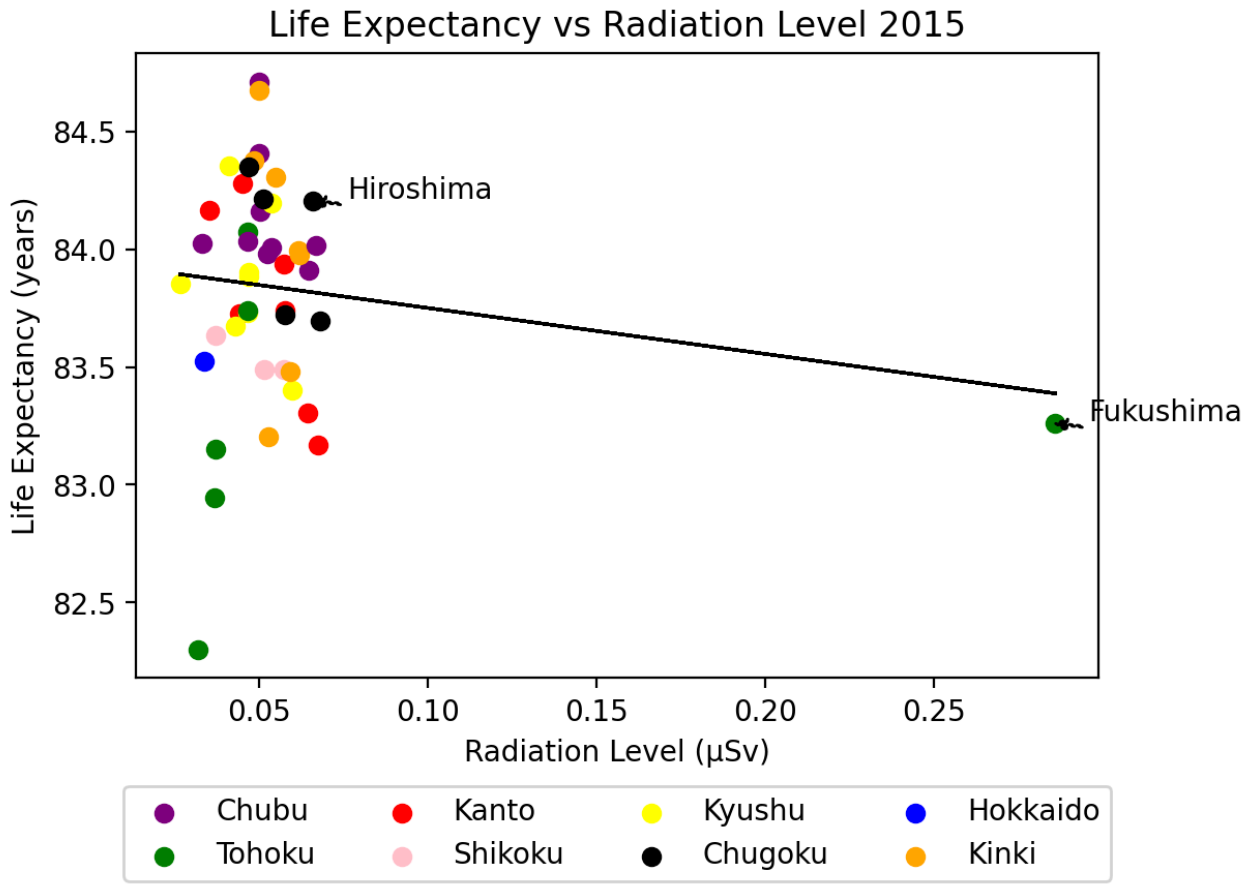


Figure 2: Demonstrates the relationship between life expectancy and radiation levels across prefectures in the year 2020. Graph is color coded by regions of Japan, while each individual dot is a prefecture. Includes a regression line and labels for specified prefectures.

The second graph, shown in Figure 2, displays the same comparison of life expectancy and radiation levels in 2015, three years after the Fukushima nuclear disaster. Every prefecture remained under 0.1 μSv except for Fukushima prefecture, which experienced a large jump to 0.29 μSv due to the nuclear meltdown at Fukushima Dai-ichi. The graph strongly displays the wide gap Fukushima had to the rest of the prefectures in regards to radiation. Additionally, the life expectancy for Fukushima increased to 83.26. Comparatively, Hiroshima's life expectancy

increased from 2010 to 2015 by about one year, as did their radiation levels by about 0.03 μSv. The correlation and significance are as follows:

Correlation coefficient: -0.1514984624268236

T-statistic: -1.0281500098898768

P-value: 0.3093741996375754

The correlation is not significant at the 0.05 level.

Once again, the correlation is not significant because the p-value was 0.31. However, it was interesting to see that the correlation was negative this year, meaning that the higher the radiation levels, the lower life expectancy was. This seems to have been caused by the large increase in radiation in Fukushima, however, it is important to understand that Fukushima is an outlier and may be skewing the data. Regardless, the correlation is still not significant, so the concrete claim that more radiation causes lower life expectancy cannot be made here, even though it can suggest it.

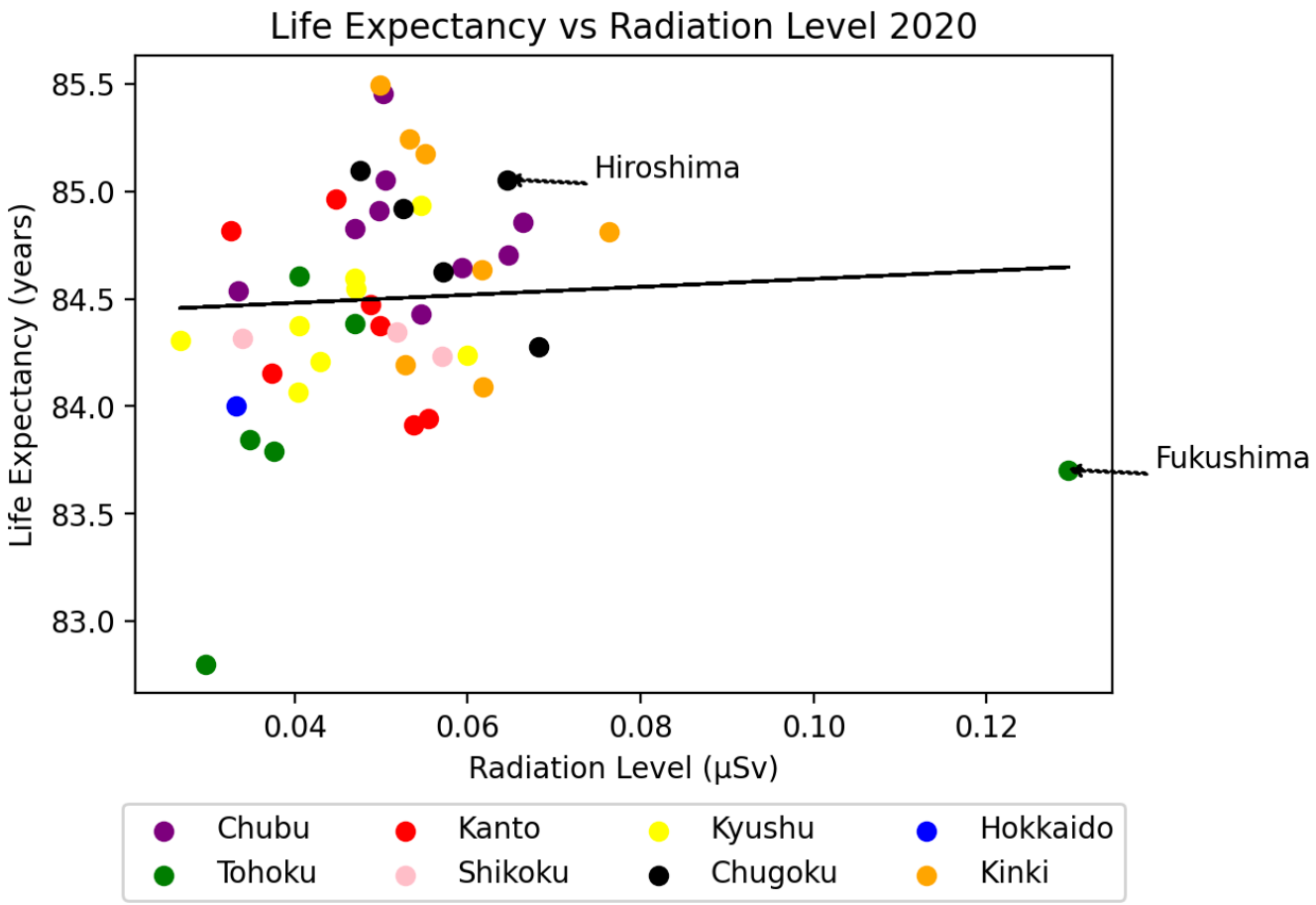


Figure 3: Demonstrates the relationship between life expectancy and radiation levels across prefectures in the year 2020. Graph is color coded by regions of Japan, while each individual dot is a prefecture. Includes a regression line and labels for specified prefectures.

Figure 3 presents the final analysis of life expectancy and radiation levels, with data collected from 2020. The main thing to notice from this graph is Fukushima's decrease of radiation levels from 0.29 μSv in 2015 to 0.13 μSv in 2020. This suggests that the cleaning up of nuclear waste and materials in the environment is working in decreasing radiation levels. Although, even nine years after the disaster the prefecture is experiencing much higher levels of radiation than other prefectures. As seen in the previous years, Hiroshima doesn't really exhibit

noticeable change. The life expectancy continued to increase and the radiation level remained stable. The correlation and significance are:

Correlation coefficient: 0.05980506087043902

T-statistic: 0.4019039222187328

P-value: 0.6896572831364236

The correlation is not significant at the 0.05 level.

Again, the data for 2020 shows a positive correlation between radiation levels and life expectancy. However, as with the previous years, the correlation is not significant. It is interesting to note that the highest radiation Fukushima experienced made the graph have a negative correlation, but as Fukushima's radiation went back down the correlation became positive again.

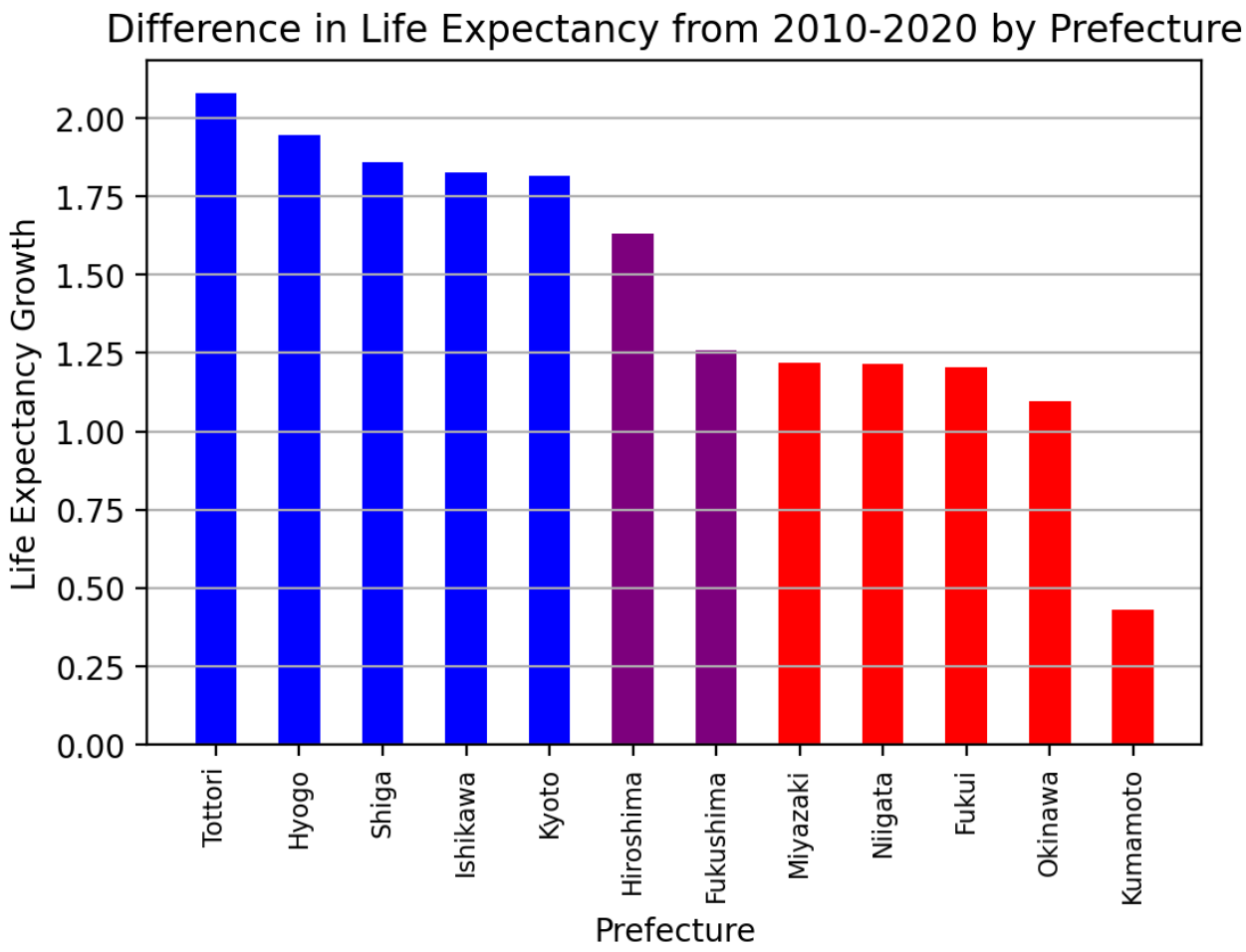


Figure 4: The difference in life expectancy from 2010-2020 in selected prefectures. Shows the top 5 prefectures with growth and the lowest 5 prefectures with the least growth, as well as Fukushima and Hiroshima.

The data shows that the life expectancy in Fukushima prefecture, even after experiencing the 3/11 disasters, has continued to increase. Figure 4 demonstrates the difference in life expectancy for a few selected prefectures from 2010 to 2020. The prefecture with the largest growth in life expectancy was Tottori, with a difference of over 2 years, and the prefecture with the least growth was Kumamoto with a difference of less than half a year. Fukushima had a relatively low growth rate in life expectancy, but it was not very low, as previously expected to see.

Findings

From my quantitative analysis of life expectancy and radiation levels, it was shown that radiation does not seem to have a significant, or any effect on life expectancy in seriously affected areas such as Hiroshima, Nagasaki, and Fukushima. It does however show that Fukushima prefecture has one of the lowest life expectancies of any prefecture, coming in with the second lowest life expectancy in 2020 at 83.71 years. But, Fukushima already exhibited one of the lowest life expectancies before the triple disasters as well, with a life expectancy of 82.45 in 2010. This was the sixth lowest life expectancy for that year. While Fukushima prefecture did experience a drop in the ratings for life expectancy, meaning it went from being the sixth lowest to the second lowest, it still remained relatively low over the 10 year period measured. The life expectancy for Fukushima prefecture went up 1.26 years, which was the 7th lowest growth in life expectancy out of the measured prefectures. The data shows that overall, Fukushima prefecture has remained pretty low on the scale of life expectancy, and considering the very large increase in radiation levels after 2011, the life expectancy measures did not reflect that. This lower life expectancy prior to the disaster seems to not have been a product of radiation exposure, as the radiation levels at the time were consistent with other prefectures.

Additionally, I took a look at radiation levels and life expectancy for Hiroshima because of the atomic bomb disaster in 1945. I thought that due to radiation exposure at the time, people might experience long-lasting negative effects and therefore die sooner than those who did not experience that. What the data showed was not necessarily the case. Hiroshima experienced an increase in life expectancy of about 3 years from 2005-2020, and 1.63 years from 2010-2020. It does not show any unusual patterns compared to other prefectures. Therefore, it would seem that

the radiation from the mid 1900's has not had a serious effect on overall life expectancy as the population in Hiroshima is living to around the same age as people in other prefectures. The same can be said about Nagasaki prefecture, another victim of atomic bombing. It has almost the same metrics as Hiroshima, with life expectancy increasing 1.61 years from 2010-2020, and radiation levels have been consistently around 0.04 μSv since 2010.

Discussion

Due to these findings, I wanted to explore why the radiation from nuclear disasters did not have any significant effect on life expectancy of those in affected areas in Fukushima prefecture. To put this into perspective, the Fukushima Dai-ichi nuclear meltdown had far less consequences than the Chernobyl disaster. Yes, Chernobyl released more than double the radiation than Fukushima, but comparatively Fukushima experienced less health issues relative to the amount of released radiation than Chernobyl did. I propose that the main reasons Fukushima prefecture residents did not exhibit a lower life expectancy than other prefectures in relation to radiation level is because of the speed of evacuation, the concerns regarding radiation from the population that kept them away from the affected area, and the increase in medical care during and after the disaster.

The World Health Organization stated that even 20 years after the Chernobyl disaster residents still lack information on how to live healthy and productive lives (World Health Organization, 2005). Additionally, the city of Chernobyl was not evacuated until 9 days after the disaster occurred (*Chernobyl*, 2024). In contrast to that, while Fukushima residents did not have an evacuation that exhibited maximum success, they still evacuated the areas much faster. The first evacuation order was issued the day following the meltdown and called for everyone in a

10km radius to evacuate. Later that day, the evacuation order was expanded to a 20km radius. (*Fukushima Prefecture*, 2019). While we now know that the evacuation area was not consistent with the spread of radiation, it still undoubtedly was a better start than no evacuation orders.

Furthermore, many residents have experienced and continue to experience immense anxiety related to living in areas affected by the nuclear meltdown. While the fear and anxiety of returning to previously evacuated areas is by no means positive for individuals in regards to mental health and wellbeing, it did however keep people away from the radiation exposed towns. Therefore, it contributed to the lack of correlation between life expectancy and radiation in Fukushima. During a discussion with Azby Brown, a lead researcher at Safecast, he discussed the internal conflict within individuals about whether to leave their homes or not. It was a tough decision for many to make, as it was very difficult to have their livelihood uprooted and figure out how to live and work in a new environment. 160,000 people evacuated the prefecture from within the 20km evacuation zone around Fukushima Dai-ichi plant, but the total number of evacuees was around 400,000 individuals, more than necessary according to the evacuation order (*ReliefWeb*, 2015). Multiple reports and sources suggest that the stress and anxiety caused by the event was a significant factor in it as well. While this paper will not explore the extent of which the stress and anxiety affected individuals, it proves to be an important factor in what kept people away from Fukushima Dai-ichi and the radiation exposed areas, and thus lowered people's radiation levels enough to not have affected life expectancy.

Finally, there was substantial medical care following the disaster. A study on medical evacuation following disasters in Japan explains that medical personnel have been receiving training on how to respond to disasters since 2002 (Okumura & Tokuno, 2015). Disaster Medical Assistance Teams (DMATS) were established for the purpose of transporting patients from

hospitals that were unsafe because of the triple disasters to hospitals in safe areas, as well as airlift individuals who were seriously injured during the disaster or could not evacuate. In the evacuation zone around Fukushima Dai-ichi 506 bedridden patients were successfully evacuated with the use of these medical teams and disaster procedures. Additionally, three months after the disaster the prefectural government established the Fukushima Health Management Survey (FHMS) with the purpose of "monitoring the long-term health of residents, promoting their well-being, and monitoring any health effects related to long-term, low-dose radiation exposure." (Tanigawa, 2018). The creation of the survey demonstrates the government's attempts to improve and/or curb health-related effects of the nuclear disaster. Finally, the literature review touched on the fact that more cancer cases arose after the disaster, but part of that is because more people have been screened consistently than ever before. Around 300,000 children participated in the voluntary thyroid screenings held by the FHMS. Furthermore, community meetings were held by Fukushima Medical University to better communicate with children and their families on the issue. Even though individuals still exhibited cases of cancer, which is not acceptable, the health measures taken afterwards were at least able to identify it and treat more people. A mix of government response, as seen with evacuation procedures, and community aid aimed at teaching community members the health implication of the disaster is ultimately how the medical response kept life expectancy from decreasing.

Limitations

One major limitation for this study is the lack of data over an extended period of time. I was only able to assess life expectancy numbers from 2010-2020. By being able to look at a wider range of life expectancy statistics I could have analyzed larger trends and also been able to

do more analysis of life expectancy and radiation in Hiroshima and Nagasaki. Because of the period of the data that I had, it was best to focus mostly on the Fukushima accident. Additionally, it is possible that not enough time has passed to accurately assess the effects of radiation on life expectancy for Fukushima prefecture. Because the accident happened 9 years before my most recent life expectancy data, it might still be reflecting numbers that are not caused by the nuclear disaster. For example, if radiation has long term effects that cause individuals to pass at 70 rather than 80, the majority of the population are not old enough yet for the data to be reflecting that. That is another reason why it would've been good to do a more in depth analysis on Hiroshima and Nagasaki, if the data had been available.

Another smaller limitation from my analysis is that I was missing data for three prefectures: Kochi, Oita, and Miyazaki, due to an error in data collection methods. While I cannot say it would have made a substantial difference, it is possible that having that data for those missing prefectures, my results could have been slightly skewed from what I actually found. Finally, I also experienced some technological constraints when working with the data. Some coding skills that I wish I had were missing, meaning it took longer to write certain code, or limitations were experienced in what kind of visualizations I was able to create.

If I were to expand this research I would look more into Hiroshima and Nagasaki and how their life expectancy levels changed because of the radiation exposure there. However, the qualitative analysis suggests that they did not experience much of a difference there either, it would still be eye opening to see quantitative data regarding it. In addition to that, it would be interesting to do more of a comparative analysis on the events of Chernobyl and Fukushima, going more in depth on the differences in the disasters and why those differences existed and ultimately saved a lot of people in the Fukushima disaster.

Another way I would expand my research in the future is to control for more variables. It would be interesting to see what factors might have caused the slightly positive correlation seen in the life expectancy and radiation chart from 2010. By comparing other factors with life expectancy it would show what could be having a big impact on it, and furthermore, controlling for those factors could result in the true relationship between radiation and life expectancy. Some other factors to be included could be, but are not limited to, improved public health, economic development, and employment levels. These factors all aim to understand one's access to medical care and services as well as their financial ability to live a healthier lifestyle, or, the country's ability to provide a healthier lifestyle to its citizens.

Conclusion

Originally, one might expect, as I did, to find that the more radiation levels in a town, the lower the town's life expectancy would be. This assumption does not go completely unbacked, as exposure to radiation can be very dangerous to human life as discussed. This was seen with the Chernobyl disaster in Ukraine and with the findings of increased cancer cases in those exposed to radiation. This paper explored the relationship between life expectancy and radiation levels in Japan, a country that experienced three nuclear disasters in the past century. The most recent, the meltdown at Fukushima Dai-ichi nuclear power plant in 2011, caused a lot of controversy and health concerns for individuals in Fukushima prefecture and country-wide.

Analyzing data on life expectancy and radiation levels across 44 out of 47 prefectures in Japan, I found the trends to not support the claim that people in prefectures with more radiation exposure would have lower life expectancies. Additionally, many other factors might have contributed to the correlation that was found, and it would be helpful to look into that more.

What the data presented was that every prefecture, from 2010 to 2020, did have an increase in life expectancy, even Fukushima prefecture which exhibited a large increase in radiation levels. Fukushima had one of the lowest life expectancies in 2020, but it has historically had a lower life expectancy. The fact that it was one of the lowest in 2020 might not mean much because of this. Also, Fukushima prefecture had an increase in life expectancy in the ten years measured that was amongst the lowest. However, those other prefectures with a life expectancy growth rate did not experience high levels of radiation, which suggests that radiation exposure does not have a large effect on life expectancy. To support that, the correlation found in all three of the measured years was not statistically significant. In future, I would like to examine other factors that play a role in life expectancy and relate to disaster recovery such as financial status and access to public health.

Finally, I found the speed of evacuation, anxiety about living in radiation exposed areas, and increased medical care to be plausible explanations for why radiation did not have a significant effect on life expectancy. Hundreds of thousands of people evacuated within the first few days of the power plant meltdown (*Fukushima Daiichi Power Plant Disaster: How Many People Were Affected? 2015 Report - Japan | ReliefWeb*, 2015). I believe this was crucial, as health issues from radiation can start as early as hours after the exposure (United States Environmental Protection Agency, 2018). The anxiety and fear faced by individuals helped them to leave in the first place, as well as keep them out of affected areas for time to come. The literature review touched on the anxiety those who evacuated faced, those who moved back to previously evacuated areas still face, and the narrative provided by the government that conditions are safe. While it is not ideal to live under that state of anxiety and have little knowledge about the effects of the disaster at the time, the qualitative analysis has shown that it played a role in keeping people safe from radiation. Lastly, medical care and aid from individuals

was able to keep the community safe during and following the disaster. The use of DMATs and successful hospital evacuations had a positive impact on the medical evacuation, and the FHMS and community meetings held to spread awareness and knowledge about the health issues individuals were facing were able to proactively address health concerns following the meltdown. To conclude, Fukushima prefecture exhibited life expectancy levels that were not affected by radiation levels caused by the Fukushima Dai-ichi nuclear power plant meltdown. However, my findings conclude that actions by individuals and communities were able to prevent and curb the lasting effects of radiation on the population in such a way that allows for no effect on the longevity of life.

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